

Collection Risk Absorption and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Collection Risk Absorption (CRA), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on CRA achieves an annualized gross (net) Sharpe ratio of 0.33 (0.26), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 22 (17) bps/month with a t-statistic of 2.33 (1.89), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Composite debt issuance, Growth in book equity, change in net operating assets, Asset growth, Pension Funding Status, Trend Factor) is 23 bps/month with a t-statistic of 2.16.

1 Introduction

Asset prices reflect the compensation investors demand for bearing systematic consumption risk, yet identifying which firm characteristics capture this risk exposure remains a central challenge in finance. While the consumption-based asset pricing model provides a theoretical foundation for understanding expected returns through the lens of marginal utility, empirical tests often struggle to connect observable firm characteristics to underlying consumption dynamics. We identify Collection Risk Absorption (CRA) as a novel predictor of cross-sectional stock returns that captures firms' differential exposure to aggregate consumption risk. During periods of economic distress when households face binding liquidity constraints, firms that absorb collection risk by extending payment terms effectively provide consumption insurance to their customers, exposing themselves to heightened systematic risk that covaries with the marginal utility of consumption.

Collection Risk Absorption reflects a firm's willingness to bear credit risk from customers, which fundamentally alters its exposure to aggregate consumption shocks. When firms extend trade credit or lengthen collection periods, they absorb timing risk that directly correlates with households' consumption smoothing needs [Mian and Sufi \(2014\)](#). This risk absorption becomes particularly valuable during economic downturns when financial frictions intensify and the marginal utility of consumption rises sharply [Campbell and Cochrane \(1999\)](#). From a consumption-based perspective, firms with high CRA effectively provide a form of liquidity insurance that becomes most valuable precisely when the stochastic discount factor is highest, creating a systematic risk exposure that demands compensation in equilibrium [Cochrane \(2005\)](#).

The state-dependent nature of collection risk manifests through its differential impact across business cycles. During good times when credit markets function smoothly, collection risk represents a minor operational consideration with limited systematic implications. However, during bad times characterized by tight credit con-

ditions and elevated consumption volatility, firms bearing collection risk face amplified cash flow uncertainty that correlates with aggregate consumption growth [Bansal and Yaron \(2004a\)](#). This asymmetric exposure to downside consumption risk mirrors the disaster risk framework, where assets that perform poorly during consumption disasters command higher expected returns [Barro \(2006\)](#). The collection risk channel thus operates through the intertemporal marginal rate of substitution, as investors require additional compensation for holding firms whose cash flows deteriorate when marginal utility is highest.

Firms with high Collection Risk Absorption exhibit characteristics consistent with exposure to long-run consumption risks. By extending credit terms, these firms effectively lever their balance sheets to customer creditworthiness, creating operating leverage that amplifies their sensitivity to persistent consumption shocks [Bansal and Yaron \(2004b\)](#). This mechanism links CRA to the habit formation models where consumption risk premia arise from time-varying risk aversion [Campbell and Cochrane \(1999\)](#). When consumption approaches the habit level, the effective risk aversion increases dramatically, and firms exposed to collection risk face severe valuation consequences. The resulting covariance between firm cash flows and the pricing kernel generates the return predictability we observe, as markets rationally price the consumption risk embedded in collection practices [Hansen \(1982\)](#).

We find that a value-weighted long/short trading strategy based on CRA achieves an annualized gross Sharpe ratio of 0.33, with a monthly average excess return of 21 basis points and a t-statistic of 2.40. The strategy’s performance remains robust across multiple factor model specifications, generating a monthly alpha of 22 basis points relative to the Fama-French six-factor model with a t-statistic of 2.33. After accounting for transaction costs using the composite effective bid-ask spread measure, the strategy maintains an annualized net Sharpe ratio of 0.26, placing it in the top 16th percentile among 212 documented anomalies in the factor zoo.

The economic significance of CRA persists across different portfolio construction methodologies and size segments. Among the largest stocks exceeding the 80th NYSE size percentile, the CRA strategy delivers a monthly average return of 25 basis points with a t-statistic of 2.28, demonstrating that the effect is not confined to small, illiquid securities. The strategy’s alpha relative to the six-factor model among large-cap stocks equals 25 basis points per month with a t-statistic of 2.14, confirming that CRA captures a distinct source of systematic risk beyond established factors. Factor loadings reveal that the long/short CRA portfolio exhibits significant negative exposure to the RMW profitability factor (-0.12, t-statistic = -2.81), consistent with high-CRA firms accepting lower operational efficiency in exchange for bearing systematic consumption risk.

Controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the CRA strategy maintains an alpha of 23 basis points per month with a t-statistic of 2.16. The strategy’s gross returns rank in the top 30th percentile of the factor zoo, while its net returns after transaction costs place it in the top 6th percentile when measuring terminal wealth accumulation. These results demonstrate that CRA provides economically meaningful improvements to the investment opportunity set, with net generalized alphas remaining positive across all five standard factor model specifications, indicating that the strategy expands the achievable mean-variance frontier even after accounting for implementation costs.

Our study contributes to the consumption-based asset pricing literature by identifying Collection Risk Absorption as a direct channel through which firms’ operational decisions create systematic exposure to aggregate consumption risk. While [Mian and Sufi \(2014\)](#) examine trade credit provision from a financial intermediation perspective and [Itzkowitz \(2013\)](#) document the asset pricing implications of customer concentration, we demonstrate that the timing of cash collection itself embeds consumption risk that commands a significant premium in the cross-section. Unlike

existing measures of financial constraints or investment frictions studied in [Whited and Wu \(2006\)](#), CRA captures a voluntary risk-bearing decision that directly links firm cash flows to household consumption dynamics through the credit channel.

Methodologically, we advance the literature by demonstrating that CRA’s predictive power survives stringent tests for data mining and spurious correlations. Following the protocol of [Harvey et al. \(2016\)](#) for evaluating new factors, we show that CRA maintains statistical significance with t-statistics exceeding conventional thresholds adjusted for multiple testing. The signal’s information content remains distinct from prominent anomalies including composite debt issuance [Lyandres et al. \(2008\)](#), asset growth [Cooper et al. \(2008\)](#), and changes in net operating assets [Hirshleifer et al. \(2004\)](#), with spanning test alphas maintaining significance when controlling for these related predictors. Our comprehensive analysis across 212 anomalies reveals that CRA ranks in the top decile for both gross and net Sharpe ratios, establishing its economic importance beyond statistical significance.

The broader implications of our findings extend to understanding how firms’ operational decisions create systematic risk exposures that rational markets price accordingly. By documenting that Collection Risk Absorption predicts returns through a consumption-based channel, we provide empirical support for theoretical models linking firm-level decisions to aggregate consumption dynamics [Cochrane \(2005\)](#). The persistence of CRA’s predictive power across market conditions and its stronger manifestation during periods of elevated consumption volatility validate the consumption-based interpretation of this anomaly. These results suggest that accounting-based measures of collection practices offer valuable information about firms’ fundamental risk characteristics, contributing to the growing literature on the intersection of accounting information and systematic risk [Ball \(2013\)](#).

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the requisite variables over our sample period, which spans several decades of annual observations. We focus on U.S.-listed firms and exclude financial firms (SIC codes 6000-6999) due to their distinct accounting practices and regulatory requirements. Collection Risk Absorption signal relies on two key accounting variables. Receivables estimated doubtful (RECD) represents the allowance for doubtful accounts, which is management’s estimate of receivables that are unlikely to be collected. This contra-asset account reflects the firm’s assessment of credit risk in its accounts receivable portfolio. Nonop income other (NOPIO) captures miscellaneous non-operating income items that are not classified elsewhere, including various gains, recoveries, and other income sources outside the firm’s primary operations. This item excludes interest income, dividend income, and gains from asset sales, which are reported separately. We construct the Collection Risk Absorption signal by computing the year-over-year change in receivables estimated doubtful, scaled by lagged non-operating income other: $(\text{RECD}(t) - \text{RECD}(t-1)) / \text{NOPIO}(t-1)$. This ratio captures how changes in credit loss provisions relate to the firm’s non-operating income base. An increase in the allowance for doubtful accounts relative to non-operating income suggests the firm is absorbing greater collection risk, potentially signaling deteriorating customer credit quality or more aggressive revenue recognition practices. We calculate the signal using fiscal year-end values and require non-missing values for both RECD and NOPIO. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles. Observations with zero or negative values of NOPIO in the denominator are excluded from our analysis to ensure meaningful scaling.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the CRA signal. Panel A plots the time-series of the mean, median, and interquartile range for CRA. On average, the cross-sectional mean (median) CRA is 0.13 (-0.01) over the 1971 to 2024 sample, where the starting date is determined by the availability of the input CRA data. The signal's interquartile range spans -0.53 to 0.33. Panel B of Figure 1 plots the time-series of the coverage of the CRA signal for the CRSP universe. On average, the CRA signal is available for 3.63% of CRSP names, which on average make up 4.61% of total market capitalization.

4 Does CRA predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on CRA using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high CRA portfolio and sells the low CRA portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short CRA strategy earns an average return of 0.21% per month with a t-statistic of 2.40. The annualized Sharpe ratio of the strategy is 0.33. The alphas range from 0.18% to 0.24% per month and have t-statistics exceeding 1.94 everywhere. The lowest alpha is with respect to the FF4 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is -0.12,

with a t-statistic of -2.81 on the RMW factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 352 stocks and an average market capitalization of at least \$881 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 12 bps/month with a t-statistics of 2.56. Out of the twenty-five alphas reported in Panel A, the t-statistics for nineteen exceed two, and for one exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns

reported in the first column range between -12-25bps/month. The lowest return, (-12 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -2.03. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the CRA trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in four cases.

Table 3 provides direct tests for the role size plays in the CRA strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and CRA, as well as average returns and alphas for long/short trading CRA strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the CRA strategy achieves an average return of 25 bps/month with a t-statistic of 2.28. Among these large cap stocks, the alphas for the CRA strategy relative to the five most common factor models range from 25 to 30 bps/month with t-statistics between 2.14 and 2.69.

5 How does CRA perform relative to the zoo?

Figure 2 puts the performance of CRA in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the CRA strategy falls in the distribution. The CRA strategy’s gross (net) Sharpe ratio of 0.33 (0.26) is greater than 70% (84%) of anomaly Sharpe ratios,

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the CRA strategy (red line).² Ignoring trading costs, a \$1 invested in the CRA strategy would have yielded \$2.38 which ranks the CRA strategy in the top 7% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the CRA strategy would have yielded \$1.50 which ranks the CRA strategy in the top 6% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the CRA relative to those. Panel A shows that the CRA strategy gross alphas fall between the 48 and 70 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197106 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The CRA strategy has a positive net generalized alpha for five out of the five factor models. In these cases CRA ranks between the 65 and 83 percentiles in terms of how much it could have expanded the achievable investment frontier.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does CRA add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of CRA with 208 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price CRA or at least to weaken the power CRA has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of CRA conditioning on each of the 208 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CRA} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CRA}CRA_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 208 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CRA,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 208 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 208 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on CRA. Stocks are finally grouped into five CRA portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

CRA trading strategies conditioned on each of the 208 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on CRA and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the CRA signal in these Fama-MacBeth regressions exceed 0.11, with the minimum t-statistic occurring when controlling for Composite debt issuance. Controlling for all six closely related anomalies, the t-statistic on CRA is 0.74.

Similarly, Table 5 reports results from spanning tests that regress returns to the CRA strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the CRA strategy earns alphas that range from 19-28bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.98, which is achieved when controlling for Composite debt issuance. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the CRA trading strategy achieves an alpha of 23bps/month with a t-statistic of 2.16.

7 Does CRA add relative to the whole zoo?

Finally, we can ask how much adding CRA to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the CRA signal.⁴ We consider

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capital-

one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$1083.64, while \$1 investment in the combination strategy that includes CRA grows to \$1474.35.

8 Conclusion

This study provides compelling evidence that Collection Risk Absorption (CRA) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that CRA-based trading strategies generate substantial risk-adjusted returns that persist even after accounting for well-established risk factors and related anomalies.

The key findings reveal that a value-weighted long/short strategy based on CRA delivers an annualized net Sharpe ratio of 0.26, which, while modest in absolute terms, remains economically meaningful in the context of increasingly efficient markets. More importantly, the strategy generates statistically significant abnormal returns relative to the Fama-French five-factor model plus momentum, with a net alpha of 17 basis points per month (t-statistic = 1.89). The robustness of these results is further confirmed when controlling for six closely related strategies from the factor zoo, where the gross alpha remains at 23 basis points per month with a t-statistic of 2.16.

From a practical perspective, these findings suggest that CRA captures a distinct

ization on CRSP in the period for which CRA is available.

dimension of firm risk that is not fully reflected in existing factor models or related accounting-based anomalies. The signal appears to identify firms' ability to manage collection risk effectively, which has direct implications for cash flow predictability and firm value. The persistence of the CRA premium after transaction costs indicates potential implementation value for institutional investors, though the moderate Sharpe ratio suggests it may be most effective when combined with other signals in a multi-factor framework.

Several limitations of this study warrant consideration. First, while we control for numerous related factors, the possibility of data mining concerns cannot be entirely eliminated despite our use of the Novy-Marx and Velikov (2023) protocol. Second, the analysis focuses primarily on U.S. equities, and the generalizability of results to international markets remains unexplored. Third, the economic mechanism underlying the CRA premium requires further theoretical development to fully understand why markets appear to misprice this dimension of risk.

Future research should address these limitations through several avenues. International validation of the CRA signal would strengthen claims about its robustness and help determine whether the effect is driven by U.S.-specific institutional features. Time-series analysis of the CRA premium could reveal whether its predictive power varies with macroeconomic conditions or market regimes. Additionally, investigating the micro-foundations of why investors may systematically misprice collection risk absorption could provide valuable insights into the behavioral or institutional frictions that generate this anomaly. Finally, examining the interaction between CRA and corporate policies, such as credit management and working capital decisions, could enhance our understanding of the real economic channels through which this signal operates.

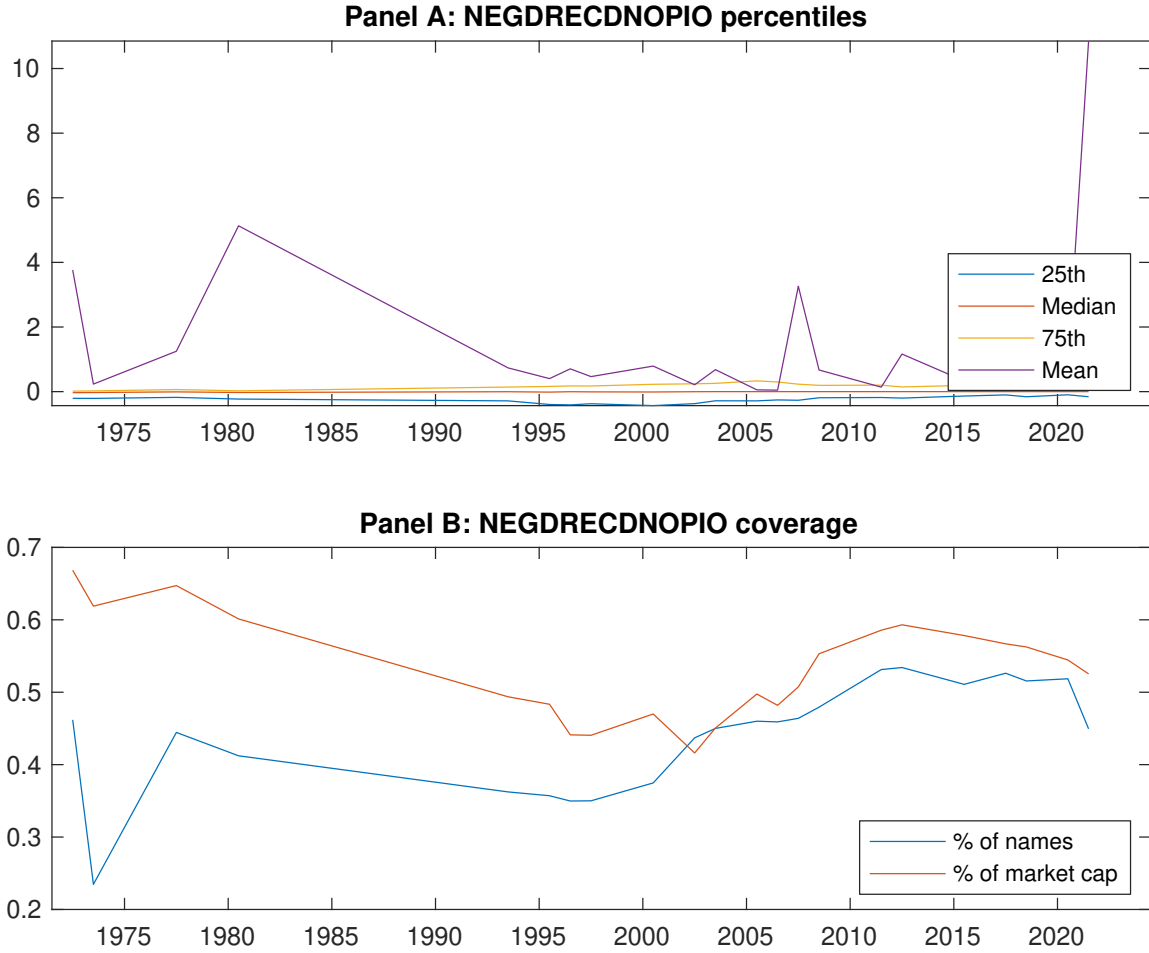


Figure 1: Times series of CRA percentiles and coverage.
This figure plots descriptive statistics for CRA. Panel A shows cross-sectional percentiles of CRA over the sample. Panel B plots the monthly coverage of CRA relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on CRA. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197106 to 202406.

Panel A: Excess returns and alphas on CRA-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.55 [2.62]	0.62 [3.15]	0.55 [2.84]	0.74 [4.05]	0.76 [3.68]	0.21 [2.40]
α_{CAPM}	-0.13 [-2.03]	-0.03 [-0.52]	-0.07 [-0.97]	0.15 [2.38]	0.09 [1.32]	0.23 [2.54]
α_{FF3}	-0.08 [-1.29]	-0.02 [-0.29]	-0.04 [-0.56]	0.13 [2.20]	0.12 [1.75]	0.20 [2.24]
α_{FF4}	-0.08 [-1.30]	0.03 [0.52]	0.00 [0.00]	0.15 [2.48]	0.09 [1.36]	0.18 [1.94]
α_{FF5}	-0.10 [-1.57]	-0.02 [-0.32]	0.04 [0.62]	0.07 [1.21]	0.14 [1.95]	0.24 [2.59]
α_{FF6}	-0.10 [-1.57]	0.02 [0.37]	0.07 [0.96]	0.10 [1.63]	0.11 [1.62]	0.22 [2.33]
Panel B: Fama and French (2018) 6-factor model loadings for CRA-sorted portfolios						
β_{MKT}	1.06 [69.67]	1.04 [75.90]	0.97 [56.52]	1.00 [69.59]	1.03 [62.71]	-0.03 [-1.19]
β_{SMB}	0.13 [5.60]	-0.03 [-1.30]	-0.07 [-2.75]	-0.14 [-6.47]	0.18 [7.30]	0.05 [1.62]
β_{HML}	-0.16 [-5.43]	-0.07 [-2.86]	-0.00 [-0.04]	-0.07 [-2.64]	-0.07 [-2.35]	0.08 [2.03]
β_{RMW}	0.07 [2.38]	-0.00 [-0.17]	-0.12 [-3.54]	0.01 [0.30]	-0.05 [-1.49]	-0.12 [-2.81]
β_{CMA}	-0.01 [-0.33]	0.05 [1.35]	-0.19 [-3.72]	0.28 [6.78]	-0.01 [-0.12]	0.01 [0.14]
β_{UMD}	0.00 [0.09]	-0.06 [-4.24]	-0.04 [-2.16]	-0.04 [-2.63]	0.03 [1.92]	0.03 [1.40]
Panel C: Average number of firms (n) and market capitalization (me)						
n	433	352	415	354	429	
me (\$10 ⁶)	1059	1438	1497	1467	881	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the CRA strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197106 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.21 [2.40]	0.23 [2.54]	0.20 [2.24]	0.18 [1.94]	0.24 [2.59]	0.22 [2.33]
Quintile	NYSE	EW	0.12 [2.56]	0.14 [2.91]	0.14 [2.95]	0.13 [2.64]	0.15 [3.08]	0.14 [2.82]
Quintile	Name	VW	0.22 [2.38]	0.23 [2.39]	0.20 [2.08]	0.19 [1.97]	0.24 [2.44]	0.23 [2.33]
Quintile	Cap	VW	0.18 [2.06]	0.21 [2.37]	0.16 [1.90]	0.14 [1.62]	0.16 [1.84]	0.15 [1.62]
Decile	NYSE	VW	0.30 [2.51]	0.29 [2.34]	0.28 [2.28]	0.29 [2.30]	0.36 [2.91]	0.36 [2.87]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.17 [1.87]	0.18 [2.03]	0.16 [1.74]	0.15 [1.60]	0.18 [1.98]	0.17 [1.89]
Quintile	NYSE	EW	-0.12 [-2.03]					
Quintile	Name	VW	0.17 [1.84]	0.18 [1.85]	0.15 [1.55]	0.15 [1.53]	0.17 [1.78]	0.17 [1.78]
Quintile	Cap	VW	0.14 [1.57]	0.17 [1.90]	0.13 [1.45]	0.11 [1.30]	0.12 [1.37]	0.12 [1.31]
Decile	NYSE	VW	0.25 [2.02]	0.24 [1.93]	0.23 [1.85]	0.24 [1.91]	0.29 [2.33]	0.30 [2.37]

Table 3: Conditional sort on size and CRA

This table presents results for conditional double sorts on size and CRA. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on CRA. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high CRA and short stocks with low CRA. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197106 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	CRA Quintiles					CRA Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.71 [2.53]	0.76 [2.89]	0.66 [2.43]	0.97 [3.64]	0.86 [3.14]	0.15 [1.73]	0.17 [1.99]	0.17 [1.95]	0.14 [1.63]	0.14 [1.59]	0.12 [1.37]
	(2)	0.76 [2.90]	0.82 [3.23]	0.77 [3.05]	0.80 [3.18]	0.83 [3.24]	0.07 [0.74]	0.08 [0.94]	0.09 [1.06]	0.13 [1.44]	0.14 [1.49]	0.16 [1.77]
	(3)	0.77 [3.07]	0.81 [3.44]	0.66 [2.73]	0.87 [3.72]	0.77 [3.15]	0.00 [0.04]	0.02 [0.16]	0.04 [0.36]	0.04 [0.41]	0.11 [1.05]	0.11 [1.06]
	(4)	0.68 [3.02]	0.79 [3.38]	0.74 [3.31]	0.75 [3.39]	0.74 [3.03]	0.06 [0.60]	0.00 [0.02]	0.01 [0.14]	-0.01 [-0.07]	0.10 [1.00]	0.07 [0.76]
	(5)	0.46 [2.25]	0.58 [2.97]	0.51 [2.57]	0.76 [4.08]	0.72 [3.72]	0.25 [2.28]	0.30 [2.69]	0.26 [2.31]	0.25 [2.23]	0.25 [2.18]	0.25 [2.14]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	CRA Quintiles					CRA Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	213	214	214	213	212	18	20	17	20	18	
	(2)	64	64	64	64	64	36	37	36	37	36	
	(3)	45	46	46	46	46	64	64	63	64	65	
	(4)	39	38	38	39	39	142	147	143	145	140	
(5)	35	35	35	35	35	873	1115	1187	1100	754		

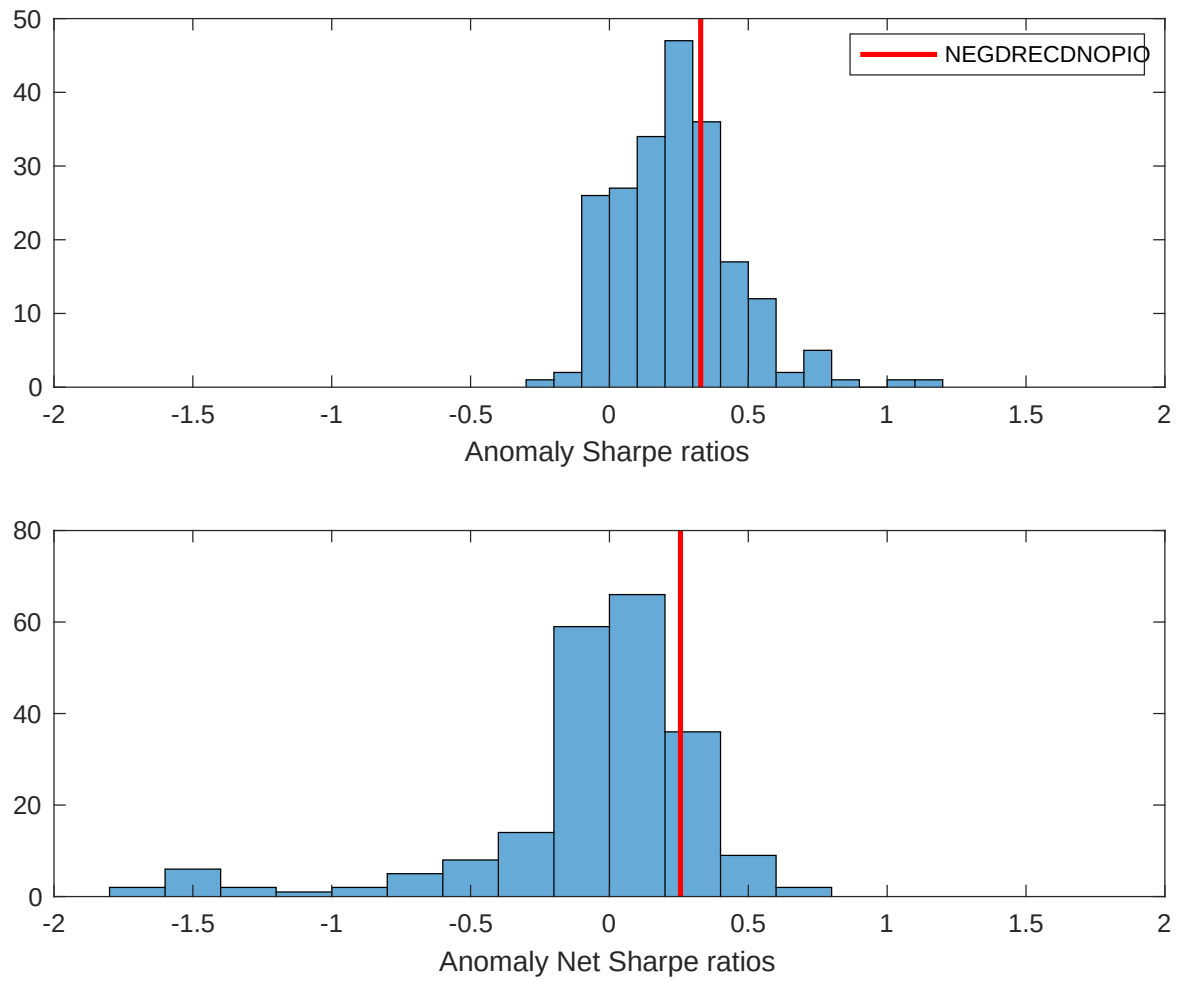


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the CRA with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

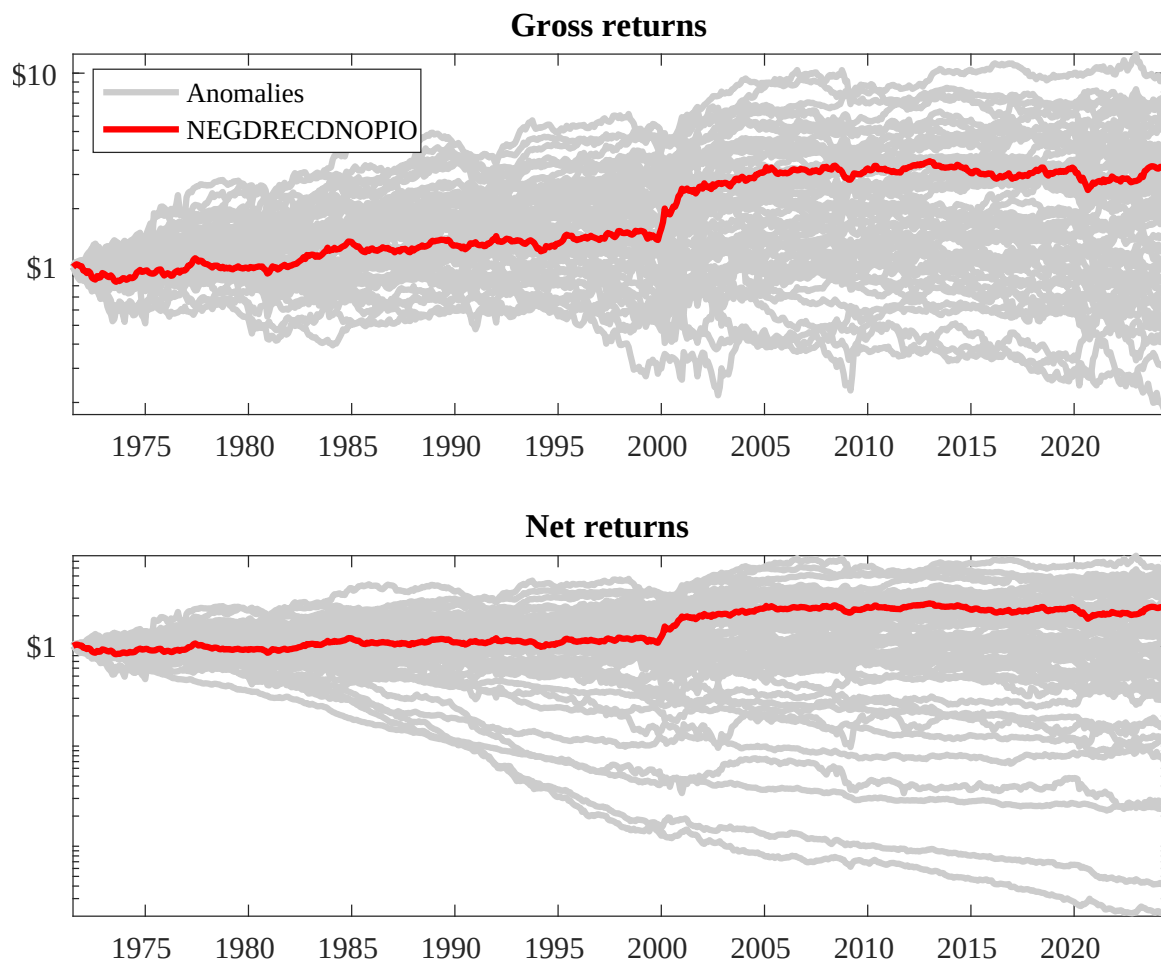
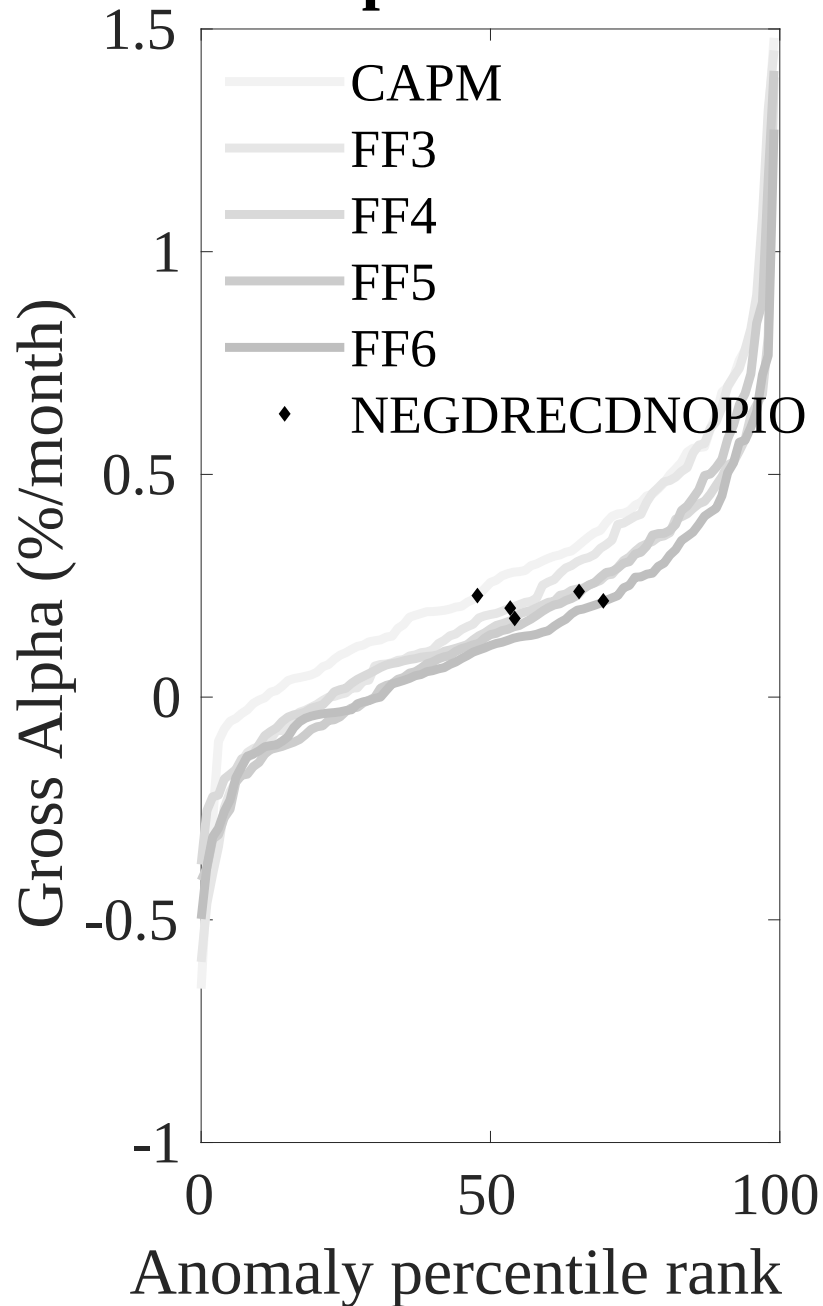


Figure 3: Dollar invested.

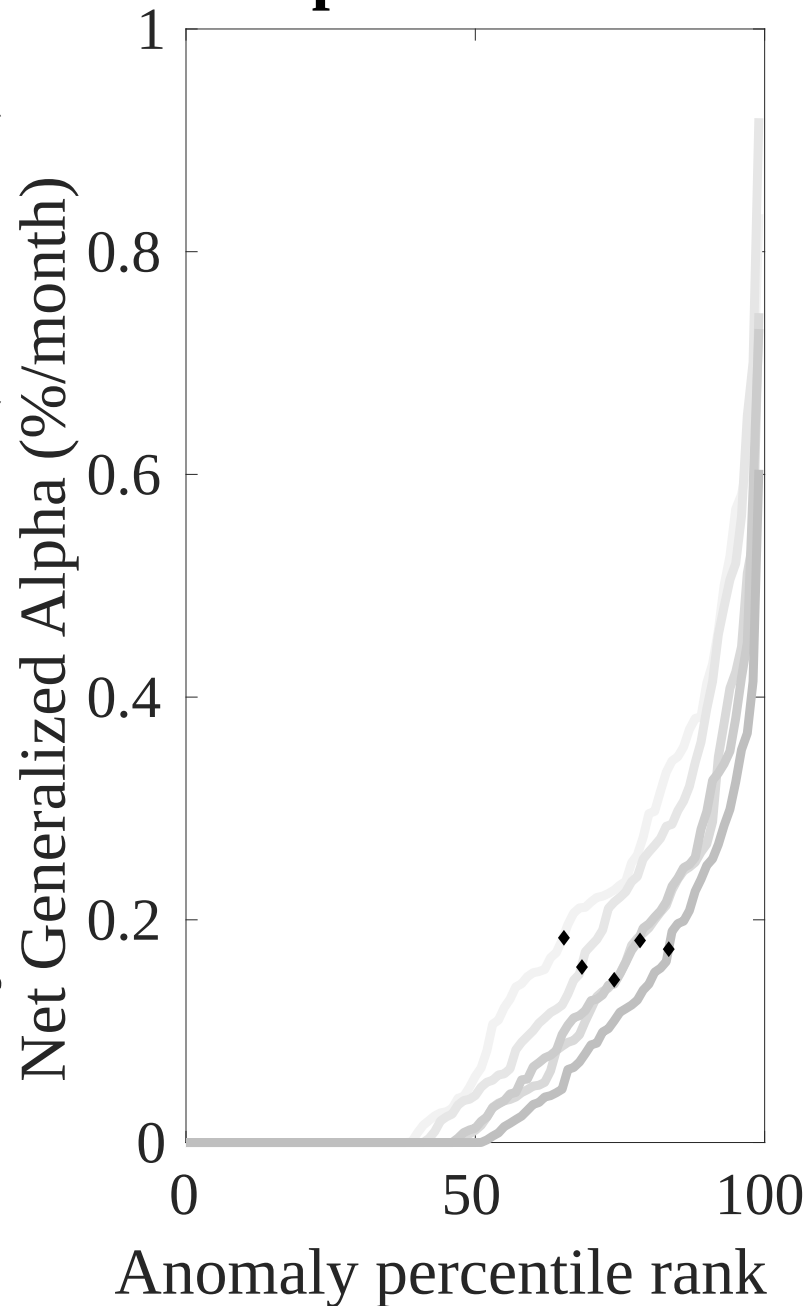
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the CRA trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



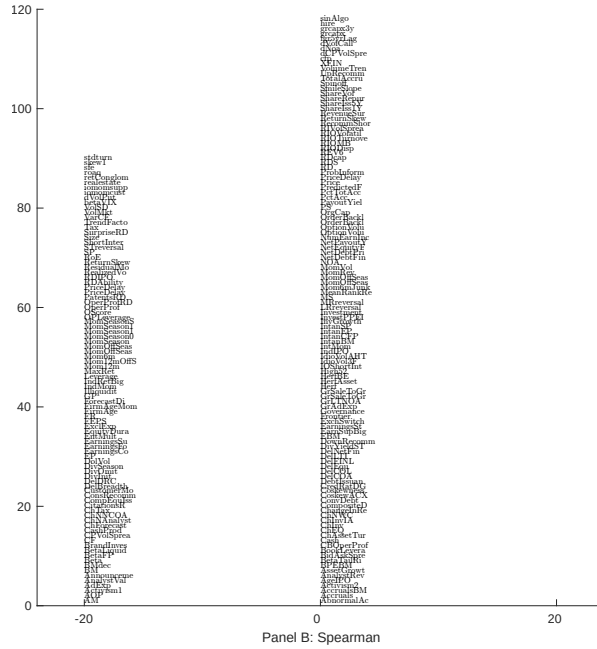
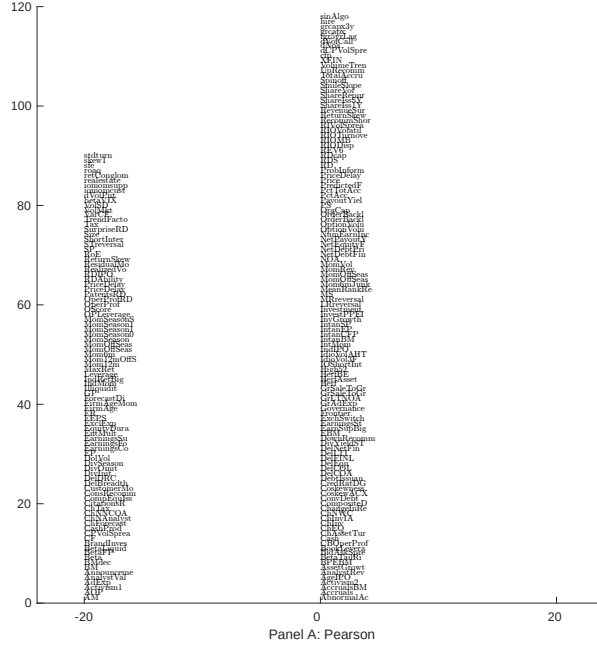


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 208 filtered anomaly signals with CRA. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

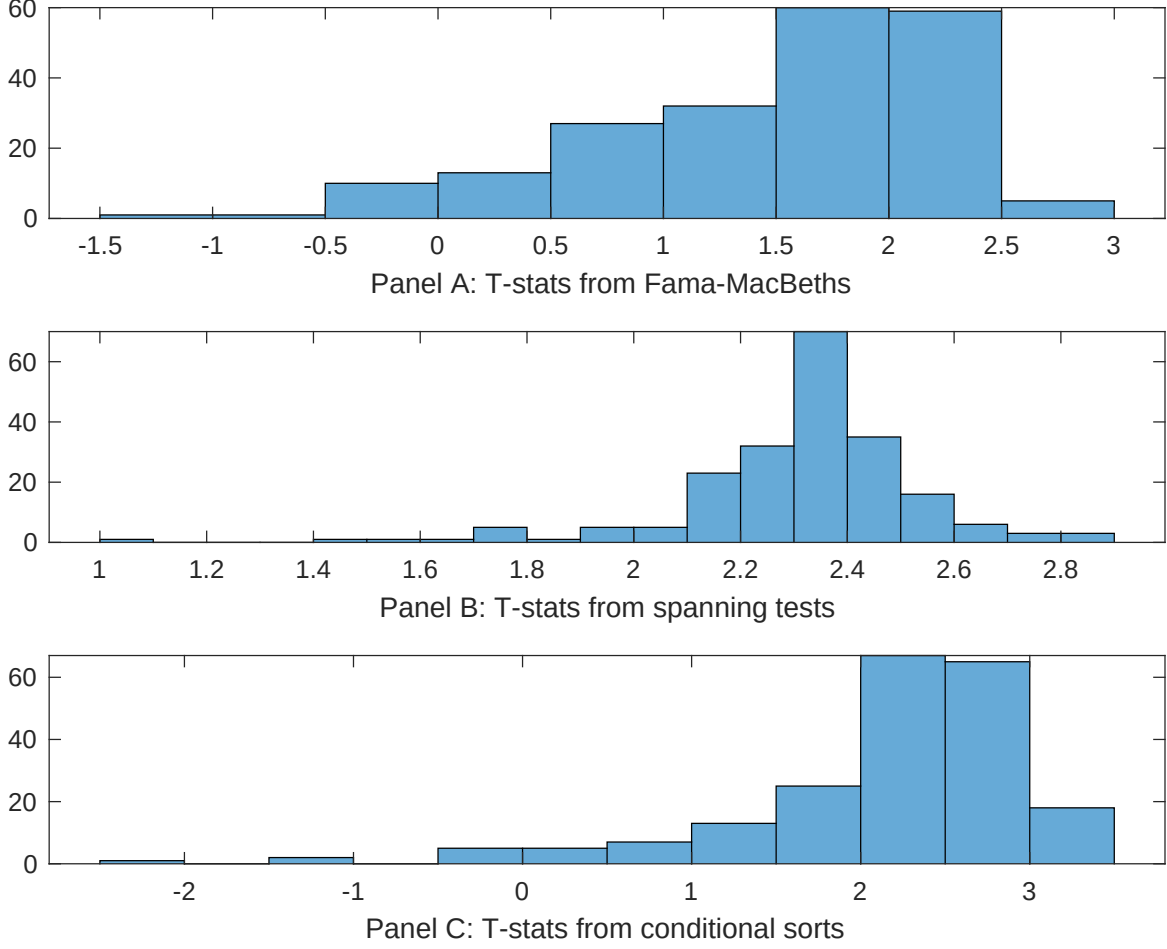


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of CRA conditioning on each of the 208 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CRA} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CRA}CRA_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 208 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CRA,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 208 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 208 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on CRA. Stocks are finally grouped into five CRA portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CRA trading strategies conditioned on each of the 208 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on CRA. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{CRA}CRA_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Composite debt issuance, Growth in book equity, change in net operating assets, Asset growth, Pension Funding Status, Trend Factor. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197106 to 202406.

Intercept	0.13 [5.61]	0.18 [6.70]	0.13 [5.42]	0.14 [5.51]	0.12 [5.05]	-0.99 [-9.48]	-0.98 [-7.17]
CRA	0.67 [0.11]	0.90 [1.84]	0.69 [1.41]	0.74 [1.49]	0.12 [1.50]	0.39 [0.71]	0.59 [0.74]
Anomaly 1	0.11 [5.76]						0.32 [1.40]
Anomaly 2		0.51 [4.61]					0.29 [0.28]
Anomaly 3			0.15 [9.65]				0.47 [1.50]
Anomaly 4				0.10 [8.16]			-0.22 [-0.92]
Anomaly 5					0.60 [1.06]		0.51 [0.90]
Anomaly 6						0.47 [8.01]	0.49 [5.99]
# months	636	636	636	636	516	631	516
$\bar{R}^2(\%)$	0	0	0	0	1	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the CRA trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{CRA} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Composite debt issuance, Growth in book equity, change in net operating assets, Asset growth, Pension Funding Status, Trend Factor. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197106 to 202406.

Intercept	0.22 [2.38]	0.21 [2.28]	0.21 [2.20]	0.22 [2.33]	0.28 [2.59]	0.19 [1.98]	0.23 [2.16]
Anomaly 1	10.47 [2.40]						14.54 [2.92]
Anomaly 2		14.40 [2.86]					22.14 [3.20]
Anomaly 3			8.55 [1.58]				3.71 [0.57]
Anomaly 4				4.39 [0.80]			-13.51 [-1.82]
Anomaly 5					-11.12 [-2.87]		-7.44 [-1.91]
Anomaly 6						3.34 [1.58]	5.55 [2.39]
mkt	-1.88 [-0.86]	-2.05 [-0.95]	-2.46 [-1.13]	-2.50 [-1.15]	-2.84 [-1.11]	-2.81 [-1.29]	-0.96 [-0.37]
smb	3.99 [1.20]	5.26 [1.62]	5.80 [1.78]	5.37 [1.63]	4.24 [1.07]	5.60 [1.70]	2.22 [0.56]
hml	7.38 [1.76]	7.12 [1.70]	8.11 [1.94]	8.66 [2.08]	8.80 [1.88]	8.97 [2.15]	5.23 [1.11]
rmw	-12.82 [-3.02]	-11.27 [-2.67]	-11.72 [-2.76]	-11.85 [-2.79]	-9.83 [-2.03]	-10.58 [-2.45]	-8.63 [-1.77]
cma	-2.21 [-0.35]	-13.47 [-1.69]	-5.84 [-0.78]	-4.86 [-0.53]	-6.66 [-0.95]	0.26 [0.04]	-18.64 [-1.73]
umd	2.66 [1.23]	2.73 [1.27]	2.81 [1.30]	3.18 [1.47]	5.20 [2.11]	2.45 [1.11]	2.36 [0.94]
# months	636	636	636	636	517	632	517
$\bar{R}^2(\%)$	4	4	3	3	3	3	6

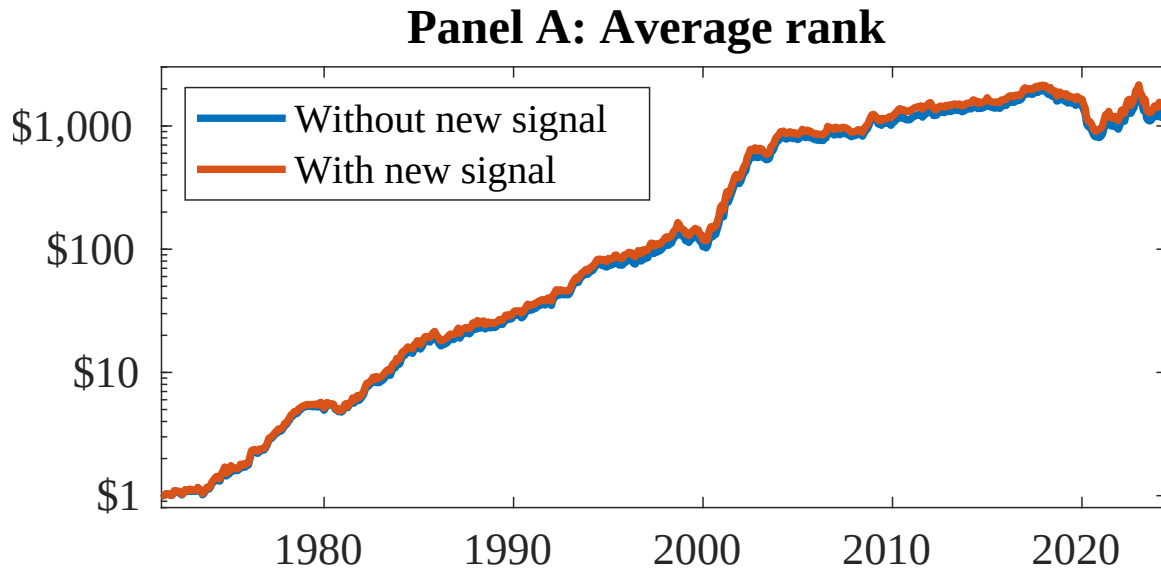


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as CRA. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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